



Qualification Pack



Junior Technician (Natural Fibre Geotextile)

QP Code: JUTE/TSC/Q4001

Version: 1.0

NSQF Level: 3

Indian Jute Industries Research Association || 17, Taratala Road, Kolkata – 700 088
West Bengal || email:pradip@ijira.org.in



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JUTE/TSC/Q4001: Junior Technician (Natural Fibre Geotextile)

Brief Job Description

Junior Technician (Natural Fibre Geotextile) should be able to manufacture product, processes and quality required at various geotextile products as per standard.

Personal Attributes

The proposed qualification is related to jute yarn production which is mostly organized and carried out in Jute Mill.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [JUTE/TSC/N4002: Raw Material for various Geotextile products](#)
2. [JUTE/TSC/N4003: Standardization of Geotextile process](#)
3. [JUTE/TSC/N4004: Machinery for construction of Geotextiles](#)
4. [JUTE/TSC/N4005: Manufacturing of Geotextiles product with desired specification](#)
5. [JUTE/TSC/N4006: Basic Maintenance of Geotextile machinery](#)
6. [JUTE/TSC/N4007: Product quality of Geotextiles](#)
7. [JUTE/TSC/N4008: Maintain health, safety and security at work place \(geotextile\)](#)
8. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Textile
Sub-Sector	Jute
Occupation	Geotextile (Natural Fibre)
Country	India
NSQF Level	3



Qualification Pack

Credits	10
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 2142.0300
Minimum Educational Qualification & Experience	10th grade pass with NA of experience No experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	1.0
Reference code on NQR	QG-03-TX-03506-2025-V1-IJIRA
NQR Version	1.0



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JUTE/TSC/N4002: Raw Material for various Geotextile products

Description

Different natural fibers Ramie/Jute/ Coir/Sisal/Hemp/Flax can be used as geotextiles for various applications

Scope

The scope covers the following :

- Understand the technical textiles, specific area of applications, and its importance

Elements and Performance Criteria

Raw Material for various Geotextile products

To be competent, the user/individual on the job must be able to:

- PC1.** Knowledge Physical properties of Fibre for Geotextiles
- PC2.** Demonstrate the Mechanical properties like a) tenacity, b) tensile strength, c) bursting strength d) drapability, e) compatibility, f) flexibility, g) tearing strength, h) frictional resistance
- PC3.** Demonstrate the Hydraulic properties such as a) porosity, b) permeability, c) permittivity, d)transitivity, e) turbidity /soil retention, f) filtration length etc
- PC4.** Explain the Degradation properties such as a) biodegradation, b) hydrolytic degradation, c) photo degradation, d) chemical degradation, e) mechanical degradation, f) other degradation occurring due to attack of rodent, termite etc.
- PC5.** Carry out Fibre Testing parameters – strength, fineness, root, defects, colour, MR% etc.
- PC6.** Demonstrate of improved Endurance properties:a) elongation b) abrasion resistance, c) clogging length and flow etc

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge Physical properties of Fibre for Geotextiles
- KU2.** Understand the Mechanical properties like a) tenacity, b) tensile strength, c) bursting strength d) drapability, e) compatibility, f) flexibility, g) tearing strength, h) frictional resistance
- KU3.** Understand the Hydraulic properties such as a) porosity, b) permeability, c) permittivity, d)transitivity, e) turbidity /soil retention, f) filtration length etc
- KU4.** Understand the Degradation properties such as a) biodegradation, b) hydrolytic degradation, c) photo degradation, d) chemical degradation, e) mechanical degradation, f) other degradation occurring due to attack of rodent, termite etc.
- KU5.** Knowledge to carry out Fibre Testing parameters like strength, fineness, root, defects, colour, MR etc.
- KU6.** Knowledge on Endurance properties like a) elongation b) abrasion resistance, c) clogging length and flow etc



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and understand written instructions
- GS2.** Communicate with supervisor appropriately
- GS3.** Write clear and short sentences in simple language for daily reporting
- GS4.** Talk to others to convey information effectively
- GS5.** Identify the real reason for the problem faced



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Raw Material for various Geotextile products</i>	25	20	-	5
PC1. Knowledge Physical properties of Fibre for Geotextiles	4	4	-	-
PC2. Demonstrate the Mechanical properties like a) tenacity, b) tensile strength, c) bursting strength d) drapability, e) compatibility, f) flexibility, g) tearing strength, h) frictional resistance	5	4	-	1
PC3. Demonstrate the Hydraulic properties such as a) porosity, b) permeability, c) permittivity, d)transitivity, e) turbidity /soil retention, f) filtration length etc	4	3	-	1
PC4. Explain the Degradation properties such as a) biodegradation, b) hydrolytic degradation, c) photo degradation, d) chemical degradation, e) mechanical degradation, f) other degradation occurring due to attack of rodent, termite etc.	4	3	-	1
PC5. Carry out Fibre Testing parameters – strength, fineness, root, defects, colour, MR% etc.	4	3	-	1
PC6. Demonstrate of improved Endurance properties:a) elongation b) abrasion resistance, c) clogging length and flow etc	4	3	-	1
NOS Total	25	20	-	5



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National Occupational Standards (NOS) Parameters

NOS Code	JUTE/TSC/N4002
NOS Name	Raw Material for various Geotextile products
Sector	Textile
Sub-Sector	
Occupation	Geotextile (Natural Fibre)
NSQF Level	3
Credits	0.5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025



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JUTE/TSC/N4003: Standardization of Geotextile process

Description

Check various machine and quality parameters following SOP

Scope

The scope covers the following :

- Illustrate the SOP of yarn & fabric manufacturing process requirements for Geotextile applications.
- Understand the process flowchart for natural fibre process (Yarn & Fabric Manufacturing)
- Design different structures in Geotextile manufacture according to various applications

Elements and Performance Criteria

Standardization of Geotextile process

To be competent, the user/individual on the job must be able to:

- PC1.** Illustrate the yarn & fabric properties and its requirements for Geotextile applications
- PC2.** Understand the mechanism of formation of yarns for different geotextiles
- PC3.** Prepare flow chart and batch composition for manufacturing of yarn
- PC4.** Understand the mechanism of formation of woven fabrics for different geotextiles
- PC5.** Prepare flow chart woven fabric quality
- PC6.** Understand the mechanism of formation of nonwoven fabrics for different geotextiles
- PC7.** Carry out the sliver / yarn quality as per quality standard of geotextile product
- PC8.** Measurement of Sliver, yarn weight at different stages
- PC9.** Carry out quality and standards required in the jute mill in Jute batching, carding, drawing and spinning section
- PC10.** Carry out Production calculations of spinning, weaving & nonwoven machineries

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Illustrate the yarn & fabric properties and its requirements for Geotextile applications
- KU2.** Understand the mechanism of formation of yarns for different geotextiles
- KU3.** Knowledge on flow chart and batch composition for manufacturing of yarn
- KU4.** Understand the mechanism of formation of woven fabrics for different geotextiles
- KU5.** Knowledge on flow chart woven fabric quality
- KU6.** Understand the mechanism of formation of nonwoven fabrics for different geotextiles
- KU7.** Knowledge to carry out the sliver / yarn quality as per quality standard of geotextile product
- KU8.** Understand measurement of Sliver, yarn weight at different stages
- KU9.** Knowledge to carry out quality and standards required in the jute mill in Jute batching, carding, drawing and spinning section



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KU10. Knowledge to carry out Production calculations of spinning, weaving & nonwoven machineries

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and understand written instructions
- GS2.** Communicate with supervisor appropriately
- GS3.** Write clear and short sentences in simple language for daily reporting
- GS4.** Talk to others to convey information effectively
- GS5.** Identify the real reason for the problem faced
- GS6.** Apply problem solving approaches in different situations
- GS7.** Refer anomalies to the supervisor



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Standardization of Geotextile process</i>	25	40	-	10
PC1. Illustrate the yarn & fabric properties and its requirements for Geotextile applications	3	4	-	1
PC2. Understand the mechanism of formation of yarns for different geotextiles	3	4	-	1
PC3. Prepare flow chart and batch composition for manufacturing of yarn	3	4	-	1
PC4. Understand the mechanism of formation of woven fabrics for different geotextiles	3	4	-	1
PC5. Prepare flow chart woven fabric quality	3	4	-	1
PC6. Understand the mechanism of formation of nonwoven fabrics for different geotextiles	2	4	-	1
PC7. Carry out the sliver / yarn quality as per quality standard of geotextile product	2	4	-	1
PC8. Measurement of Sliver, yarn weight at different stages	2	4	-	1
PC9. Carry out quality and standards required in the jute mill in Jute batching, carding, drawing and spinning section	2	4	-	1
PC10. Carry out Production calculations of spinning, weaving & nonwoven machineries	2	4	-	1
NOS Total	25	40	-	10



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National Occupational Standards (NOS) Parameters

NOS Code	JUTE/TSC/N4003
NOS Name	Standardization of Geotextile process
Sector	Textile
Sub-Sector	
Occupation	Geotextile (Natural Fibre)
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025



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JUTE/TSC/N4004: Machinery for construction of Geotextiles

Description

Identification of machine parts & machine overview

Scope

The scope covers the following :

- Identifying critical machine parts and its functions and operate the Yarn and Fabric manufacturing machinery
- Production operation in yarn and fabric manufacturing machinery.
- Explain various quality used in industry like Geo-textile fabric, Lino fabric, conventional fabric etc

Elements and Performance Criteria

Machinery for construction of Geotextiles

To be competent, the user/individual on the job must be able to:

- PC1.** Explain Jute Spreader / Softener, Carding, drawing and spinning machine and identification of parts
- PC2.** Calculate Draft & doubling and its necessity for sliver and yarn preparation
- PC3.** Carry out by hand and eye method the following quality parameters of raw jute fibres during selection, Strength, Root content, Defects, Colour, Fineness, Bulk-Density
- PC4.** Prepare quality wise Batch mix
- PC5.** Carryout the moisture content of jute morah has been taken properly for future reference
- PC6.** Perform basic maintenance work to be carried out in the shift in case of necessity
- PC7.** Carry out starting/stopping/control switches are O.K. in case individual drive and alternative arrangements in case of group drive
- PC8.** Knowledge on machine safety arrangement functioning correctly, in case of jam in the feed roller
- PC9.** Carry out proper working of Spreader / Softener, Carding and drawing machine and uniformity of the sliver otherwise take corrective measure
- PC10.** Carry out grade/quality wise (with proper colour coding) stocking of doffed spreader /carded roles at appropriate place for further processing
- PC11.** Maintaining proper sliver path for ensuring perfect drawing (also checking of stop motion)
- PC12.** Calculate wastages like sliver, fleece, droppings time to time and take weight to understand the corrective measures required
- PC13.** Coordinate with maintenance people for proper setting of draft change pinion, twist pinion, felt bob position etc. for different count of yarn
- PC14.** Identify jute sliver quality and jute yarn quality according to desired quality standards and specifications for the geotextile product
- PC15.** Find out the causes of end break in spinning machine, end break in winding (warp and weft) machine and take the remedial measures for excessive end breakage



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- PC16.** Follow the instruction of Superior /shift In-charge during change of count and quality and Check frequently the quality of cop and spool and ensure that quality standards and specifications are maintained
- PC17.** Explain Winding / Beaming machine and identification of parts
- PC18.** Explain Sizing and its necessity and measurement of machine performance at weaving preparatory section
- PC19.** Carry out correct take up of sizing paste by the yarn and squeezing properly, adjust the pressing roller accordingly
- PC20.** Carry out moisture content (30%-35% for sackings & 18% -22% for hessian) & 1.5-2 % size application in beam in consultation with superiors as practiced in jute industry and ensure 70%. wet pick-up of size
- PC21.** Carry out the weavers knot is of minimum size to pass easily through the heald wires and reed
- PC22.** Carry out that all the stop motions, indication lamps , preventive mechanisms etc., function properly
- PC23.** Explain Loom Motions and its classification, Identification and function of required tools, equipment, and consumables needed for the Production
- PC24.** Correct the fabric defects like wrong drawing, wrong denting, end out , double end etc., immediately and also ensure that the other fabric defects too are corrected at the earliest, before continuing further production
- PC25.** Rectify in case of any floats, should be able to " unweave " the same & should be able to run the machine without " starting mark or crack"

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand Jute Spreader / Softener, Carding, drawing and spinning machine and identification of parts
- KU2.** Knowledge on Calculation of Draft & doubling and its necessity for sliver and yarn preparation
- KU3.** Knowledge to carry out by hand and eye method the following quality parameters of raw jute fibres during selection, Strength, Root content, Defects, Colour, Fineness, Bulk-Density
- KU4.** Understand quality wise Batch mix
- KU5.** Understand moisture content of jute morah
- KU6.** Understand basic maintenance work to be carried out in the shift in case of necessity
- KU7.** Understand starting/stopping/control switches are O.K. in case individual drive and alternative arrangements in case of group drive
- KU8.** Knowledge on machine safety arrangement functioning correctly, in case of jam in the feed roller
- KU9.** Carry out proper working of Spreader / Softener, Carding and drawing machine and uniformity of the sliver otherwise take corrective measure
- KU10.** Understand grade/quality wise (with proper colour coding) stocking of doffed spreader /carded roles at appropriate place for further processing



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- KU11.** Knowledge on Maintaining proper sliver path for ensuring perfect drawing (also checking of stop motion)
- KU12.** Understand Calculation wastages like sliver, fleece, droppings time to time and take weight to understand the corrective measures required
- KU13.** Coordinate with maintenance people for proper setting of draft change pinion, twist pinion, felt bob position etc. for different count of yarn
- KU14.** Identify jute sliver quality and jute yarn quality according to desired quality standards and specifications for the geotextile product
- KU15.** Understand the causes of end break in spinning machine, end break in winding (warp and weft) machine and take the remedial measures for excessive end breakage
- KU16.** Knowledge on quality and Check frequently the quality of cop and spool and ensure that quality standards and specifications are maintained
- KU17.** Knowledge Winding / Beaming machine and identification of parts
- KU18.** Knowledge on Sizing and its necessity and measurement of machine performance at weaving preparatory section
- KU19.** Understand to find out correct take up of sizing paste by the yarn and squeezing properly, adjust the pressing roller accordingly
- KU20.** Knowledge to carry out moisture content (30-35 percent for sackings & 18 -22 percent for hessian) & 1.5-2 percent size application in beam in consultation with superiors as practiced in jute industry and ensure 70 percent. wet pick-up of size
- KU21.** Knowledge to carry out the weavers knot is of minimum size to pass easily through the heald wires and reed
- KU22.** Knowledge to carry out that all the stop motions, indication lamps , preventive mechanisms etc., function properly
- KU23.** Understand Loom Motions and its classification
- KU24.** Identification and function of required tools, equipment, and consumables needed for the Production
- KU25.** Knowledge on fabric defects like wrong drawing, wrong denting, end out , double end etc., immediately and also ensure that the other fabric defects too are corrected at the earliest, before continuing further production

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and understand written instructions
- GS2.** Communicate with supervisor appropriately
- GS3.** Write clear and short sentences in simple language for daily reporting
- GS4.** Identify the real reason for the problem faced
- GS5.** Refer anomalies to the supervisor
- GS6.** Seek clarification on problems from others



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Machinery for construction of Geotextiles</i>	25	40	-	10
PC1. Explain Jute Spreader / Softener, Carding, drawing and spinning machine and identification of parts	1	2	-	-
PC2. Calculate Draft & doubling and its necessity for sliver and yarn preparation	1	2	-	-
PC3. Carry out by hand and eye method the following quality parameters of raw jute fibres during selection, Strength, Root content, Defects, Colour, Fineness, Bulk-Density	1	2	-	-
PC4. Prepare quality wise Batch mix	1	2	-	-
PC5. Carryout the moisture content of jute morah has been taken properly for future reference	1	2	-	-
PC6. Perform basic maintenance work to be carried out in the shift in case of necessity	1	2	-	-
PC7. Carry out starting/stopping/control switches are O.K. in case individual drive and alternative arrangements in case of group drive	1	2	-	-
PC8. Knowledge on machine safety arrangement functioning correctly, in case of jam in the feed roller	1	2	-	-
PC9. Carry out proper working of Spreader / Softener, Carding and drawing machine and uniformity of the sliver otherwise take corrective measure	1	2	-	-
PC10. Carry out grade/quality wise (with proper colour coding) stocking of doffed spreader /carded roles at appropriate place for further processing	1	2	-	-
PC11. Maintaining proper sliver path for ensuring perfect drawing (also checking of stop motion)	1	2	-	1
PC12. Calculate wastages like sliver, fleece, droppings time to time and take weight to understand the corrective measures required	1	2	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Coordinate with maintenance people for proper setting of draft change pinion, twist pinion, felt bob position etc. for different count of yarn	1	2	-	1
PC14. Identify jute sliver quality and jute yarn quality according to desired quality standards and specifications for the geotextile product	1	2	-	1
PC15. Find out the causes of end break in spinning machine, end break in winding (warp and weft) machine and take the remedial measures for excessive end breakage	1	2	-	1
PC16. Follow the instruction of Superior /shift In-charge during change of count and quality and Check frequently the quality of cop and spool and ensure that quality standards and specifications are maintained	1	1	-	1
PC17. Explain Winding / Beaming machine and identification of parts	1	1	-	1
PC18. Explain Sizing and its necessity and measurement of machine performance at weaving preparatory section	1	1	-	1
PC19. Carry out correct take up of sizing paste by the yarn and squeezing properly, adjust the pressing roller accordingly	1	1	-	1
PC20. Carry out moisture content (30%-35% for sackings & 18% -22% for hessian) & 1.5-2 % size application in beam in consultation with superiors as practiced in jute industry and ensure 70%. wet pick-up of size	1	1	-	1
PC21. Carry out the weavers knot is of minimum size to pass easily through the heald wires and reed	1	1	-	-
PC22. Carry out that all the stop motions, indication lamps , preventive mechanisms etc., function properly	1	1	-	-
PC23. Explain Loom Motions and its classification, Identification and function of required tools, equipment, and consumables needed for the Production	1	1	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. Correct the fabric defects like wrong drawing, wrong denting, end out , double end etc., immediately and also ensure that the other fabric defects too are corrected at the earliest, before continuing further production	1	1	-	-
PC25. Rectify in case of any floats, should be able to " unweave " the same & should be able to run the machine without " starting mark or crack"	1	1	-	-
NOS Total	25	40	-	10



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National Occupational Standards (NOS) Parameters

NOS Code	JUTE/TSC/N4004
NOS Name	Machinery for construction of Geotextiles
Sector	Textile
Sub-Sector	
Occupation	Geotextile (Natural Fibre)
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025



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JUTE/TSC/N4005: Manufacturing of Geotextiles product with desired specification

Description

Elucidate various workshop machinery, basic maintenance tools used in Geotextile processing machinery

Scope

The scope covers the following :

- Perform basic workshop operations using suitable tools for fitting, riveting, milling, lathe, grinding, files, drilling etc. and Produce components by different operations and check accuracy using appropriate measuring instruments

Elements and Performance Criteria

Manufacturing of Geotextiles product with desired specification

To be competent, the user/individual on the job must be able to:

- PC1.** Knowledge of all technical terms and definition, Testing stds.,
- PC2.** Calculate MR%, Idea of RH%, Relation of RH% with MR%, QR. Calculation
- PC3.** Identify Statistical Quality Control for geotextile processing & preparation of control charts
- PC4.** Carry out various sampling technique for testing and sample collection from shop floor
- PC5.** Identify Testing standard and SOP of test method
- PC6.** Identify and demonstrate the Soil Basics
- PC7.** Identify and demonstrate the Soil testing such as Grain Size distribution, Atterberg Limit Test, CBR Test etc.
- PC8.** Carry out frequent checking of in-process materials and ensure quality management of delivered sliver, yarn and fabric during spinning, weaving & nonwoven machineries
- PC9.** : Identify techniques for methods of process control
- PC10.** Identify & rectification of any deviation found in shop floor

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of all technical terms and definition, Testing stds.,
- KU2.** Understand Calculation of MR, Idea of RH, Relation of RH with MR, QR. Calculation
- KU3.** Identify Statistical Quality Control for geotextile processing & preparation of control charts
- KU4.** Understand various sampling technique for testing and sample collection from shop floor
- KU5.** Identify Testing standard and SOP of test method
- KU6.** Understand and demonstrate the Soil Basics
- KU7.** Identify and demonstrate the Soil testing such as Grain Size distribution, Atterberg Limit Test, CBR Test etc.



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- KU8.** Knowledge to carry out frequent checking of in-process materials and ensure quality management of delivered sliver, yarn and fabric during spinning, weaving & nonwoven machineries
- KU9.** Identify techniques for methods of process control
- KU10.** Identify & rectification of any deviation found in shop floor

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and understand written instructions
- GS2.** Communicate with supervisor appropriately
- GS3.** Identify the real reason for the problem faced
- GS4.** Apply problem solving approaches in different situations
- GS5.** Refer anomalies to the supervisor
- GS6.** Seek clarification on problems from others
- GS7.** Apply good attention to detail
- GS8.** Maintain precision of work



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manufacturing of Geotextiles product with desired specification</i>	25	40	-	10
PC1. Knowledge of all technical terms and definition, Testing stds.,	3	4	-	1
PC2. Calculate MR%, Idea of RH%, Relation of RH% with MR%, QR. Calculation	3	4	-	1
PC3. Identify Statistical Quality Control for geotextile processing & preparation of control charts	3	4	-	1
PC4. Carry out various sampling technique for testing and sample collection from shop floor	3	4	-	1
PC5. Identify Testing standard and SOP of test method	3	4	-	1
PC6. Identify and demonstrate the Soil Basics	2	4	-	1
PC7. Identify and demonstrate the Soil testing such as Grain Size distribution, Atterberg Limit Test, CBR Test etc.	2	4	-	1
PC8. Carry out frequent checking of in-process materials and ensure quality management of delivered sliver, yarn and fabric during spinning, weaving & nonwoven machineries	2	4	-	1
PC9. : Identify techniques for methods of process control	2	4	-	1
PC10. Identify & rectification of any deviation found in shop floor	2	4	-	1
NOS Total	25	40	-	10



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National Occupational Standards (NOS) Parameters

NOS Code	JUTE/TSC/N4005
NOS Name	Manufacturing of Geotextiles product with desired specification
Sector	Textile
Sub-Sector	
Occupation	Geotextile (Natural Fibre)
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025



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JUTE/TSC/N4006: Basic Maintenance of Geotextile machinery

Description

Plan for standard maintenance of machineries as recommended by machine manufacturer

Scope

The scope covers the following :

- Perform basic maintenance and use relevant tools & instruments and repair / replacements of machine parts by using workshop machinery

Elements and Performance Criteria

Basic maintenance of Geotextile machinery

To be competent, the user/individual on the job must be able to:

- PC1.** Describe the working principles of workshop machineries viz. lathe, shaping machine, drilling machine, milling machine and grinding machine
- PC2.** Identify, troubleshoot and rectify different faults in the machine and machine parts.
- PC3.** Repair, overhaul and assemble the parts in the machine with the help of blueprint.
- PC4.** Identify various types of gears- spur gear, bevel gear, helical gear, worm and worm wheel, rack and pinion and its calculations
- PC5.** Identify different types of gear used in production machinery and calculate power transmission takes place through gear
- PC6.** Explain different types of belt used in production machinery and their speed transmission takes place
- PC7.** Calculate Speed ratio through different types of gear and belt drive
- PC8.** Identify different bearing in jute machinery and their application
- PC9.** List out the consumables used for maintenance such as oil, lubrication and their purpose and types
- PC10.** Identifications of machine faults at different and its rectification
- PC11.** Dismantle the electrical system to replace faulty modules and perform soldering, crimping, harnessing to secure cables and components
- PC12.** Dismantle the mechanical equipment for maintenance and assemble them
- PC13.** Lubricate the equipment using the right amount of lubricant suggested in the manual
- PC14.** Avoid waste, ascertain unused materials and components for disposal, store the seinan environmentally appropriate manner and prepare for disposal
- PC15.** Spreader Maintenance – assembly of gill bars, causes of bar jam
- PC16.** Carding Maintenance – repining and gauging and pin diameter checking, roll former diameter setting
- PC17.** Drawing Maintenance – repining, carriage setting and pressure adjustment of pressing roller
- PC18.** Spinning Maintenance – Dial gauge, height gauge



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- PC19.** Different components – Compressor, Pressure Gauge, Filter Regulator Lubricator, Valves and Actuators
- PC20.** Replace oil or grease as specified in maintenance procedure
- PC21.** Re-assemble the equipment to working condition after completing maintenance
- PC22.** Follow the document procedure replaced component, consumables used, other details of maintenance for any reference
- PC23.** Carry out maintenance activities of all equipment at recommended frequency
- PC24.** Run the equipment and test its functionality in the presence of the concerned authority
- PC25.** Carry out Checking of Beaming machine – Leveling, Starch roller condition, Loom - Catcher pin setting, Damping - Nozzle and water flow, Paper bowl condition, Needle looper and spreader setting
- PC26.** Different components – Compressor, Pressure Gauge, Filter Regulator Lubricator, Valves and Actuators
- PC27.** Re-assemble the equipment to working condition after completing maintenance
- PC28.** Follow the document procedure replaced component, consumables used, other details of maintenance for any reference
- PC29.** Run the equipment and test its functionality in the presence of the concerned authority

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the working principles of workshop machineries viz. lathe, shaping machine, drilling machine, milling machine and grinding machine
- KU2.** Identify, troubleshoot and rectify different faults in the machine and machine parts.
- KU3.** Knowledge on repair, overhaul and assemble the parts in the machine with the help of blueprint.
- KU4.** Identify various types of gears- spur gear, bevel gear, helical gear, worm and worm wheel, rack and pinion and its calculations
- KU5.** Identify different types of gear used in production machinery and calculate power transmission takes place through gear
- KU6.** Knowledge on different types of belt used in production machinery and their speed transmission takes place
- KU7.** Knowledge on Calculation of Speed ratio through different types of gear and belt drive
- KU8.** Identify different bearing in jute machinery and their application
- KU9.** Understand the consumables used for maintenance such as oil, lubrication and their purpose and types
- KU10.** Knowledge of machine faults at different and its rectification
- KU11.** Knowledge on the electrical system to replace faulty modules and perform soldering, crimping, harnessing to secure cables and components
- KU12.** Knowledge on the mechanical equipment for maintenance and assemble them
- KU13.** Knowledge on the Lubricate the equipment using the right amount of lubricant suggested in the manual



Qualification Pack

- KU14.** Understand waste, ascertain unused materials and components for disposal, store the environmentally appropriate manner and prepare for disposal.
- KU15.** Understand Spreader Maintenance assembly of gill bars, causes of bar jam
- KU16.** Understand Carding Maintenance repining and gauging and pin diameter checking, roll former diameter setting
- KU17.** Understand Drawing Maintenance repining, carriage setting and pressure adjustment of pressing roller.
- KU18.** Understand Different components Compressor, Pressure Gauge, Filter Regulator Lubricator, Valves and Actuators
- KU19.** Knowledge on replace oil or grease as specified in maintenance procedure
- KU20.** Understand replace oil or grease as specified in maintenance procedure
- KU21.** Knowledge on the document procedure replaced component, consumables used, other details of maintenance for any reference
- KU22.** Knowledge to carry out maintenance activities of all equipment at recommended frequency
- KU23.** Knowledge to run the equipment and test its functionality in the presence of the concerned authority
- KU24.** Knowledge to carry out Checking of Beaming machine Leveling, Starch roller condition, Loom - Catcher pin setting, Damping - Nozzle and water flow, Paper bowl condition, Needle looper and spreader setting
- KU25.** Understand different components Compressor, Pressure Gauge, Filter Regulator Lubricator, Valves and Actuators
- KU26.** Knowledge on re-assemble the equipment to working condition after completing maintenance
- KU27.** Knowledge on the document procedure replaced component, consumables used, other details of maintenance for any reference
- KU28.** Knowledge to run the equipment and test its functionality in the presence of the concerned authority

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and understand written instructions
- GS2.** Communicate with supervisor appropriately
- GS3.** Write clear and short sentences in simple language for daily reporting
- GS4.** Talk to others to convey information effectively
- GS5.** Identify the real reason for the problem faced
- GS6.** Apply problem solving approaches in different situations



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basic maintenance of Geotextile machinery</i>	25	40	-	10
PC1. Describe the working principles of workshop machineries viz. lathe, shaping machine, drilling machine, milling machine and grinding machine	1	2	-	-
PC2. Identify, troubleshoot and rectify different faults in the machine and machine parts.	1	2	-	-
PC3. Repair, overhaul and assemble the parts in the machine with the help of blueprint.	1	2	-	-
PC4. Identify various types of gears- spur gear, bevel gear, helical gear, worm and worm wheel, rack and pinion and its calculations	1	2	-	-
PC5. Identify different types of gear used in production machinery and calculate power transmission takes place through gear	1	2	-	-
PC6. Explain different types of belt used in production machinery and their speed transmission takes place	1	2	-	-
PC7. Calculate Speed ratio through different types of gear and belt drive	1	2	-	-
PC8. Identify different bearing in jute machinery and their application	1	2	-	-
PC9. List out the consumables used for maintenance such as oil, lubrication and their purpose and types	1	2	-	-
PC10. Identifications of machine faults at different and its rectification	1	2	-	-
PC11. Dismantle the electrical system to replace faulty modules and perform soldering, crimping, harnessing to secure cables and components	1	2	-	-
PC12. Dismantle the mechanical equipment for maintenance and assemble them	1	1	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Lubricate the equipment using the right amount of lubricant suggested in the manual	1	1	-	-
PC14. Avoid waste, ascertain unused materials and components for disposal, store the seinan environmentally appropriate manner and prepare for disposal	1	1	-	-
PC15. Spreader Maintenance – assembly of gill bars, causes of bar jam	1	1	-	-
PC16. Carding Maintenance – repining and gauging and pin diameter checking, roll former diameter setting	1	1	-	-
PC17. Drawing Maintenance – repining, carriage setting and pressure adjustment of pressing roller	1	1	-	-
PC18. Spinning Maintenance – Dial gauge, height gauge	1	1	-	-
PC19. Different components – Compressor, Pressure Gauge, Filter Regulator Lubricator, Valves and Actuators	1	1	-	-
PC20. Replace oil or grease as specified in maintenance procedure	1	1	-	1
PC21. Re-assemble the equipment to working condition after completing maintenance	1	1	-	1
PC22. Follow the document procedure replaced component, consumables used, other details of maintenance for any reference	1	1	-	1
PC23. Carry out maintenance activities of all equipment at recommended frequency	1	1	-	1
PC24. Run the equipment and test its functionality in the presence of the concerned authority	1	1	-	1
PC25. Carry out Checking of Beaming machine – Leveling, Starch roller condition, Loom - Catcher pin setting, Damping - Nozzle and water flow, Paper bowl condition, Needle looper and spreader setting	1	1	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. Different components - Compressor, Pressure Gauge, Filter Regulator Lubricator, Valves and Actuators	-	1	-	1
PC27. Re-assemble the equipment to working condition after completing maintenance	-	1	-	1
PC28. Follow the document procedure replaced component, consumables used, other details of maintenance for any reference	-	1	-	1
PC29. Run the equipment and test its functionality in the presence of the concerned authority	-	1	-	1
NOS Total	25	40	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	JUTE/TSC/N4006
NOS Name	Basic Maintenance of Geotextile machinery
Sector	Textile
Sub-Sector	
Occupation	Geotextile (Natural Fibre)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025



Qualification Pack

JUTE/TSC/N4007: Product quality of Geotextiles

Description

Knowledge on various quality parameters following SOP in Yarn & Fabric Manufacturing Process of Geotextile products

Scope

The scope covers the following :

- Explain various quality used in natural fibre based Geotextile and Skilling on process and quality control in jute spinning & weaving process

Elements and Performance Criteria

Product quality of Geotextiles

To be competent, the user/individual on the job must be able to:

- PC1.** Explain application & Installation Guidelines of Natural fibre based Geotextile in Roads
- PC2.** Explain application, & Installation Guidelines of Natural fibre based Geotextiles for Slope Protection
- PC3.** Explain application, & Installation Guidelines of Natural fibre based Geotextiles for River Bank Protection
- PC4.** Explain application & Installation Guidelines of Natural fibre based Geotextile in Road Construction
- PC5.** Understand various methods and devices that have been successfully used to monitor and evaluate geotextiles performance in the field.
- PC6.** Identify all types of geotextiles, and monitoring for all types of applications, the reinforcement area using geotextiles and geogrids is the most advanced.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge on application & Installation Guidelines of Natural fibre based Geotextile in Roads
- KU2.** Knowledge on application & Installation Guidelines of Natural fibre based Geotextile in Roads
- KU3.** Knowledge on application & Installation Guidelines of Natural fibre based Geotextile in Roads
- KU4.** Knowledge on application & Installation Guidelines of Natural fibre based Geotextile in Roads
- KU5.** Understand various methods and devices that have been successfully used to monitor and evaluate geotextiles performance in the field
- KU6.** Understand all types of geotextiles, and monitoring for all types of applications, the reinforcement area using geotextiles and geogrids is the most advanced.

Generic Skills (GS)



Qualification Pack

User/individual on the job needs to know how to:

- GS1.** Read and understand written instructions
- GS2.** Communicate with supervisor appropriately
- GS3.** Write clear and short sentences in simple language for daily reporting
- GS4.** Talk to others to convey information effectively
- GS5.** Apply good attention to detail
- GS6.** Maintain precision of work



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Product quality of Geotextiles</i>	25	20	-	5
PC1. Explain application & Installation Guidelines of Natural fibre based Geotextile in Roads	4	3	-	1
PC2. Explain application, & Installation Guidelines of Natural fibre based Geotextiles for Slope Protection	4	3	-	1
PC3. Explain application, & Installation Guidelines of Natural fibre based Geotextiles for River Bank Protection	4	3	-	1
PC4. Explain application & Installation Guidelines of Natural fibre based Geotextile in Road Construction	4	3	-	1
PC5. Understand various methods and devices that have been successfully used to monitor and evaluate geotextiles performance in the field.	4	4	-	1
PC6. Identify all types of geotextiles, and monitoring for all types of applications, the reinforcement area using geotextiles and geogrids is the most advanced.	5	4	-	-
NOS Total	25	20	-	5



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	JUTE/TSC/N4007
NOS Name	Product quality of Geotextiles
Sector	Textile
Sub-Sector	
Occupation	Geotextile (Natural Fibre)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025



Qualification Pack

JUTE/TSC/N4008: Maintain health, safety and security at work place (geotextile)

Description

Maintain health, safety and security at work place

Scope

The scope covers the following :

- Comply with health and safety related instructions applicable to the workplace

Elements and Performance Criteria

Maintain health, safety and security at work place for Geotextile applications

To be competent, the user/individual on the job must be able to:

- PC1.** Comply with health and safety related instructions applicable to the workplace
- PC2.** Use and maintain personal protective equipment such as “ear plug”, “ nose mask”, “head cap” etc., as per protocol
- PC3.** Carry out own activities in line with approved guidelines and procedures
- PC4.** Safely handle and remove waste
- PC5.** Follow environment management system related procedure
- PC6.** Identify and correct (if possible) malfunctions in machinery and equipment
- PC7.** Store materials and equipment in line with organizational requirements
- PC8.** Minimize health and safety risks to self and others due to own actions
- PC9.** Communicate the safety plan to everyone and attach disciplinary rules with the implementation
- PC10.** Monitor the workplace and work processes for potential risks and threat
- PC11.** Undertake first aid, fire-fighting and emergency response training, if asked to do so
- PC12.** Take action based on instructions in the event of fire, emergencies or accidents
- PC13.** Follow organization procedures for shutdown and evacuation when required
- PC14.** Recognise other possible security issues existing in the workplace

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand health and safety related instructions applicable to the workplace
- KU2.** Knowledge on use and maintain personal protective equipment such as ear plug, nose mask, head cap etc., as per protocol
- KU3.** Knowledge on use and maintain personal protective equipment such as ear plug, nose mask, head cap etc., as per protocol



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- KU4.** Knowledge on use and maintain personal protective equipment such as ear plug, nose mask, head cap etc., as per protocol
- KU5.** Knowledge on use and maintain personal protective equipment such as ear plug, nose mask, head cap etc., as per protocol
- KU6.** Identify and correct (if possible) malfunctions in machinery and equipment
- KU7.** Knowledge on use and maintain personal protective equipment such as ear plug, nose mask, head cap etc., as per protocol
- KU8.** Knowledge on use and maintain personal protective equipment such as ear plug, nose mask, head cap etc., as per protocol
- KU9.** Knowledge on use and maintain personal protective equipment such as ear plug, nose mask, head cap etc., as per protocol
- KU10.** Knowledge on use and maintain personal protective equipment such as ear plug, nose mask, head cap etc., as per protocol
- KU11.** Understand first aid, fire-fighting and emergency response training, if asked to do so
- KU12.** Knowledge on action, based on instructions in the event of fire, emergencies or accidents
- KU13.** Understand organization procedures for shutdown and evacuation when required
- KU14.** Understand other possible security issues existing in the workplace

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and understand written instructions
- GS2.** Communicate with supervisor appropriately
- GS3.** Identify the real reason for the problem faced
- GS4.** Apply good attention to detail
- GS5.** Maintain precision of work



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain health, safety and security at work place for Geotextile applications</i>	25	20	-	5
PC1. Comply with health and safety related instructions applicable to the workplace	2	2	-	-
PC2. Use and maintain personal protective equipment such as “ear plug”, “ nose mask”, “head cap” etc., as per protocol	2	2	-	-
PC3. Carry out own activities in line with approved guidelines and procedures	2	2	-	-
PC4. Safely handle and remove waste	2	2	-	-
PC5. Follow environment management system related procedure	2	2	-	-
PC6. Identify and correct (if possible) malfunctions in machinery and equipment	2	2	-	-
PC7. Store materials and equipment in line with organizational requirements	2	1	-	-
PC8. Minimize health and safety risks to self and others due to own actions	2	1	-	-
PC9. Communicate the safety plan to everyone and attach disciplinary rules with the implementation	2	1	-	-
PC10. Monitor the workplace and work processes for potential risks and threat	2	1	-	1
PC11. Undertake first aid, fire-fighting and emergency response training, if asked to do so	2	1	-	1
PC12. Take action based on instructions in the event of fire, emergencies or accidents	1	1	-	1
PC13. Follow organization procedures for shutdown and evacuation when required	1	1	-	1
PC14. Recognise other possible security issues existing in the workplace	1	1	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	25	20	-	5



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	JUTE/TSC/N4008
NOS Name	Maintain health, safety and security at work place (geotextile)
Sector	Textile
Sub-Sector	
Occupation	Geotextile (Natural Fibre)
NSQF Level	3
Credits	0.5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025



Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team



Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services



Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to IJIRA for conducting the assessment online through Portal with two probable dates for Assessment
- IJIRA deploys the ToA certified Assessor for executing the assessment
- IJIRA monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then Assessment will spill over to consecutive days.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

3. Assessment Quality Assurance levels/Framework:



Qualification Pack

- Question bank is created by the Subject Matter Experts (SME) are verified by the other SME
- Questions are mapped to the specified assessment criteria
- Assessor must be ToA certified.

4. Types of evidence or evidence-gathering protocol:

- reporting of the assessor from assessment location
- Centre photographs with signboards and scheme specific branding

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Method for assessment documentation, archiving, and access

- Hard copies of the documents are stored

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
JUTE/TSC/N4002.Raw Material for various Geotextile products	25	20	-	5	50	10
JUTE/TSC/N4003.Standardization of Geotextile process	25	40	-	10	75	15
JUTE/TSC/N4004.Machinery for construction of Geotextiles	25	40	-	10	75	15
JUTE/TSC/N4005.Manufacturing of Geotextiles product with desired specification	25	40	-	10	75	15
JUTE/TSC/N4006.Basic Maintenance of Geotextile machinery	25	40	-	10	75	15



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
JUTE/TSC/N4007.Product quality of Geotextiles	25	20	-	5	50	10
JUTE/TSC/N4008.Maintain health, safety and security at work place (geotextile)	25	20	-	5	50	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	195	250	-	55	500	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.